

KenaBecha JU

Core Features

The Jahangirnagar University student marketplace

A buy/sell/shop platform for JU students: personal listings, seller-run shops, real-time chat, ratings, an AI shopping assistant, and a full admin/moderation layer.

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1 Accounts & Authentication

- **JU-verified signup** — Email/password signup collects real student-verification fields — student ID, registration number, hall, department, session — with batch computed server-side and non-editable.
- **OTP email verification** — 6-digit one-time code, 10-minute expiry, attempt-limited; login is blocked until the account is verified.
- **Google one-click sign-in** — Identity Services ID-token flow, verified server-side with no client secret. Signing in with an email that already has a local account links the Google identity rather than creating a duplicate.
- **Profile completion gate** — Selling (creating a shop or listing) requires a complete JU profile; an incomplete one is routed to a re-submittable completion form instead of a one-time block.
- **Session security** — Short-lived JWT access tokens with rotating refresh tokens (httpOnly cookies); reuse of an already-rotated refresh token revokes the whole session family. Password reset and email-verification tokens are single-use.
- **Roles** — User, moderator, and admin, with moderators reaching moderation surfaces and admins additionally controlling users, roles, and site content. First-admin bootstrap via an ADMIN_EMAILS allowlist, no database shell required.

2 Listings

- **Personal and shop-based listings** — Any student can list an item personally, or attach it to their own shop.
- **Rich listing data** — Title, description, price (fixed, negotiable, or free), condition, category, tags, up to several photos with drag-and-drop reordering, pickup and/or delivery fulfillment.
- **Seller-controlled status** — Mark sold, mark out-of-stock / available again (works for personal listings too, not just shop inventory), pause/reactivate, and relist a sold or paused item with a refreshed expiry.
- **Manual shop ordering** — A shop owner can set the display order of their own listings via drag-and-drop; a shop's storefront defaults to that order.
- **Restock requests** — A buyer can ask to be notified when an out-of-stock shop listing comes back; the seller sees a count (not a name list), and every requester is notified automatically the moment the seller marks it available again.
- **Automatic lifecycle** — Listings expire and get swept automatically; a seller can renew one, and admins can feature a listing for a time-boxed promotional boost or mark one as a permanent, curated "Top" pick.
- **Related listings** — A detail page surfaces other items from the same shop, then shared tags, then recent listings — never an empty rail.

3 Shops

- **Shop storefronts** — Public shop pages with branding (logo, cover photo, description, category), listings, reviews, and post-purchase review prompts.
- **Shop dashboard** — Sellers manage their own shop's branding, details, and inventory — including a shop-scoped, reorderable listing view — from one screen.
- **Following** — Buyers can follow a shop; the shop sees a follower count.
- **Multiple shops per seller** — A student can run more than one shop, and choose which shop (or "personal") a new listing belongs to when creating it.

4 Shop Posts

- **Product-anchored posts** — A shop composes a title and a formatted description — bold, italic, underline, font, and colour on selected text — and links any number of its own listings, so a shop can announce or re-announce the same product to grab attention as often as it likes.
- **Rich text, safely stored** — Formatting is authored in a real WYSIWYG editor and sanitized server-side against a strict tag/attribute allow-list before it's ever stored, with a client-side pass as a second layer right before display.
- **Follow-ranked feed** — A site-wide feed of published posts, shops a visitor follows shown first; posts also appear on the shop's own storefront under a dedicated tab.
- **Moderation & notifications** — Every post goes through the same pending/published/rejected queue — and the same bulk-moderation and audit-log machinery — as listings and shops; publishing notifies the shop's followers, rejection tells the owner why.
- **Edits are re-reviewed** — Changing an already-published or rejected post resubmits it to pending, so a moderation decision can never be silently invalidated by a later edit.

5 Search & Discovery

- **Full-text search** — Matches listing titles, descriptions, and tags; safe against SQL wildcard injection and resilient to typos via trigram matching.
- **Autocomplete** — Live search suggestions as a visitor types, feeding straight into the browse page's filtered results.
- **Filters and sort** — Category (hierarchical, two-level taxonomy), price range, condition, tags, and sort by newest, price, or popularity.
- **Categories** — Admin-curated category tree with icons, browsable from the navigation and homepage, and reflected in listing counts.
- **View counts & trending** — Per-listing view tracking (de-duplicated per viewer) powers a "most viewed" rail and admin analytics.

6 Chat & Communication

- **Real-time messaging** — WebSocket-pushed live delivery with REST-persisted history, reliable across multiple backend worker processes via Redis pub/sub fan-out.
- **Per-listing conversations** — Each conversation is anchored to a specific listing between a buyer and seller; re-contacting reuses the same thread.
- **Owner inbox** — Unified inbox with shop filter tabs and unread badges, so a seller running several shops sees everything in one place.
- **Offline notification** — A message sent while the recipient is offline triggers an email, so nothing gets missed.

7 Ratings & Trust

- **Post-sale reviews** — A buyer can rate a shop or a seller once a listing is sold or out-of-stock, they messaged about it, and they haven't already rated it — a real-transaction gate, not a free-for-all.
- **Discoverable from either side** — The shop page's Reviews tab points a buyer straight at anything of theirs still eligible to rate, instead of requiring them to already know which listing to revisit.
- **Star ratings with review text** — 1–5 stars plus an optional written review, rolled up into an average shown on shop and profile pages.

8 AI Shopping Assistant

- **Floating chat widget** — Available site-wide, bilingual, asks a visitor what they're looking for and finds real matches.
- **Grounded in the real catalog** — Answers are produced via live tool calls against the actual listings database — the assistant can never recommend a listing a human browsing the site wouldn't also see, and nothing is ever invented.
- **Structural safety guarantee** — Only listing ids that genuinely came back from a real search during that conversation are ever shown, enforced in code rather than left to the model's own restraint.
- **Real-time streaming** — Replies stream token-by-token; the frontend renders prose as it arrives and resolves recommended listing cards once the model has finished searching.
- **Bilingual, colloquial-aware search** — Understands Bangla, Banglish, and regional food-market vocabulary (e.g. that "mutton" means goat meat locally), retrying with alternate terms rather than giving up on a literal miss.
- **Admin-tunable** — An admin can turn the assistant on/off and edit its system prompt from the dashboard, without a deploy.
- **Session-only history** — Conversation history lives in the browser's session storage only — never tied to an account or persisted server-side.

9 Notifications

- **In-app notifications** — Live-pushed via WebSocket plus a REST-backed list with read/unread state and an unread-count badge.
- **Email notifications** — New messages (when offline), new ratings, listing/shop moderation outcomes, restock availability, password reset, and welcome emails — all through one notification service.

10 Admin & Moderation

- **Reporting queue** — Any user, listing, or shop can be reported; reports feed a single admin queue resolvable as dismissed, removed, warned, or banned — with real side effects (soft-deleting content, deactivating and revoking sessions on a ban).
- **Bulk moderation** — Select many items and act on all of them at once, routed through the exact same per-item logic (and audit trail, and notifications) as acting on one — never a shortcut bulk UPDATE.
- **Admin dashboard** — Headline metrics with a switchable time window, a daily activity chart, and a most-viewed-listings rail.
- **Append-only audit log** — Every privileged action — role changes, removals, category/navigation edits, bans — is recorded with who, what, and when.
- **Navigation & content management** — Navbar and footer links, category taxonomy, modular landing-page sections, and site branding (logo, contact info, socials) are all admin-editable without a deploy.
- **Scheduled announcements** — A site-wide banner with a scheduling window, tone (info/warning/critical), and per-wording dismissal so a reader isn't re-nagged by an unrelated edit.
- **Moderator role** — A lighter-weight role for day-to-day moderation, distinct from full admin control over users, roles, and site configuration.

11 Localization & Accessibility

- **Full English/Bangla i18n** — Every user-facing string, including number and price formatting, in both languages with a persistent locale switch.
- **Responsive, accessible design** — A cohesive glass/emerald visual system with dark/light themes, keyboard navigation, skip links, and reduced-motion support; verified and fixed for real phone-width layouts (navbar, notification panel, mobile bottom navigation).
- **SEO** — Canonical URLs, Open Graph tags, sitemap, and proper error boundaries/404 handling.

12 Platform & Reliability

- **Rate limiting** — Sliding-window, IP-based limits (correctly identifying real visitor IPs behind the production proxy) on every abuse-prone endpoint — auth flows, messaging, and more.

- **Automated backups** — Nightly database backups retained on a rolling window, independent of the application containers.
- **CI/CD** — Automated backend and frontend test suites and type-checks on every push, with production deploys handled by the hosting platform's own pull-based pipeline.
- **Test coverage** — A real-Postgres backend test suite (hundreds of tests) and frontend test coverage, exercising eligibility rules, guards, and edge cases directly rather than mocking them away.
- **Soft deletion everywhere it matters** — Users, shops, and listings are soft-deleted, keeping ratings, reports, and chat history displayable after removal instead of leaving broken references.